

Contingent vs. Robust AI Power Demand — v2

From the supply side to the load side: how much announced data-centre demand survives the grid operators' own screens, in two regions

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Status: v2. Adds the demand/load side to the v0.1 supply-side anchor (Zenodo 10.5281/zenodo.20706509). Methods precedent: the bot-energy paper (Zenodo 10.5281/zenodo.20562696).

What is new in v2. v0.1 measured the *supply* side — how much announced generation-interconnection capacity actually reaches commercial operation in PJM ($\approx 22\%$ of announced MW). v2 adds the *load* side: how much announced data-centre demand survives the grid operators' own vetting, across two regions (PJM and ERCOT). The honest finding up front: on the load side the operators do the deflation themselves and publish the method, so this note does not deflate the headline — it packages the operators' own published deflation into a reproducible, cross-region, motive-neutral form, and reports that two independent operators converge on roughly half surviving the first credible screen. The supply-side sections (§1–§6) are carried from v0.1 with figures unchanged (re-run and confirmed 2026-06-15); the load-side sections (§7–§11) are new.

Abstract

Forecasts of AI-driven electricity demand treat announced capacity as if it were firm, on both sides of the meter: the generation announced to serve the load, and the data-centre load itself. This note deflates both, using the grid operators' own resolved data. On the supply side (carried from v0.1), PJM's resolved serial queue (8,253 generation projects, ≈ 837 GW announced) shows only ≈ 16 – 25% of announced MW reach commercial operation — 22.8% on the mature 2010–2017 cohort. On the load side (new in v2), the deflation is performed and published by the operators themselves: PJM has already cut its 2030 large-load request from ≈ 60 GW submitted to ≈ 38 GW accepted, of which only the ≈ 32 GW that is firm (backed by an Electric Service Obligation or Construction Commitment) counts toward capacity; its documented worked example survives at $\approx 48\%$ of requested capacity. ERCOT, whose ≈ 226 GW queue (≈ 70 – 77% data centres) was maximally padded by a near-zero barrier to entry, publishes a realized per-site de-rate of 49.8% (actual peak consumption $\div$ requested MW) and has energized 5,302 MW to date ($\approx 2\%$ of its tracking). The cross-region result: two independent operators, using a forward de-rate (PJM) and a realized one (ERCOT), converge on roughly half of requested data-centre demand surviving the first credible screen, and both now gate their forecast on an executed agreement. The contribution is not the deflation but its packaging — a reproducible, cross-region, motive-neutral referee — plus a two-rail energisation bracket that places the operators' accepted figure against a realized-completion floor. The output is a band with its softest input named, letting a planner treat any announced headline as a scenario, not a baseline.

1. The question

The same error the bot-energy paper deflated — a real headline number multiplied against a *contaminated set* and quoted as a single figure — recurs, larger, in AI power-demand projections, on both sides of the meter. On the supply side the contaminated set is the generation-interconnection queue (duplicate filings, speculative projects, projects that withdraw once costs are known). On the load side it is the data-centre interconnection queue, inflated by the same mechanisms plus an absence of standardised entry requirements. Two questions follow. Of announced *generation* capacity, how much is deliverable? And of announced data-centre *load*, how much is real? v0.1 answered the first; v2 adds the second, and finds the operators already answer it themselves.

2. Data and method

Supply side. PJM "Serial Service Request Status" (the legacy serial queue, projects entering before the July-2023 cluster transition), exported as XLSX. PJM planning data carry a redistribution restriction; this work publishes derived aggregates only, and `reproduce.py` reads a locally-downloaded file rather than bundling it. Reproducibility caveat: because the input cannot be redistributed, the supply-side script is reproducible only by someone who has independently obtained PJM access.

Load side. Three PJM documents and a set of ERCOT filings, all public:

- PJM, *Load Adjustment Request Implementation* (Resource Adequacy Planning, 1 July 2025) — the method and the worked example.
- PJM Load Analysis Subcommittee, *Load Adjustment Requests Summary for the 2026 Load Forecast — Preliminary* (24 November 2025) — the submitted / proposed / firm curves.
- PJM, *2026 Long-Term Load Forecast Report* (posted 14 January 2026) — the final accepted figures and revision magnitudes.
- ERCOT: *System Planning and Weatherization Update* (December 2025), *Large Load Interconnection Process Q&A* (24 December 2025), *Long-Term Load Forecast 2025* (the 49.8% per-site figure), and the NPRR1234 / PGRR115 market notices (the forecast gate, effective 15 December 2025).

The load-side figures are published aggregates; they require no redistribution-restricted file and are reproducible from the documents above.

The conditional-funnel discipline (supply side). Because the same projects move through queue milestones, the survival rate at each gate is conditional on the prior gate's survivors, so the rates multiply validly:

```
deliverable = headline × r2(viability) × r3(build)
r2 = MW reaching an executed Interconnection Agreement ÷ MW entered
r3 = MW In Service ÷ MW reaching an executed IA
```

$r2 \times r3 = \text{built} \div \text{entered}$ by definition — it is the decomposition restated as an identity, not an independent consistency check. De-duplication $r1$ is treated separately (§5).

3. Supply-side result (carried from v0.1)

Realized completion by cohort (MW-Energy weighted; In-Service-only):

Submitted cohort	% resolved	% built	% withdrawn
≤2009	100%	16%	84%
2010–14	100%	25%	75%
2015–17	92%	20%	71%
2018–20	75%	3% (still building)	72%
2021–25	84%	0% (too new)	84%

The mature cohorts converge on a realized completion of ≈16–25% of announced MW — roughly one announced MW in five is delivered. This is directionally corroborated by, not strictly like-for-like with, the published anchors: LBNL *Queued Up* reports ≈14–20% nationally (vs PJM-only here), and Exelon discloses ≈22% of its development pipeline (a company pipeline, not a queue-completion rate).

Funnel decomposition (2010–2017 cohort, 204 GW entered):

Stage	% of entered
Entered queue	100%
Reached Feasibility	93%
Reached System Impact	65%
Reached IA-executed	39% (= r2)
Built (In Service)	22.8%

with $r3$ (IA → service) = 58%, and $r2 \times r3 = 22.8\% = \text{built/entered}$ (the identity). 61% of announced MW exit before a signed agreement; the steepest drop is between Feasibility and the System Impact Study, where each project's share of network-upgrade costs is revealed. A further 13% exit after a signed agreement. Switching to a MW-Capacity basis moves the headline only to 24.8% — the one-in-five anchor is stable to the basis choice.

4. The named load-bearing assumption — and why the proxy holds

Cohort comparability is the supply-side assumption: the mature 2010–2017 generation is generic (wind/solar/gas) used as a completion base rate, not capacity announced to serve AI, and transferring that rate to the AI-era cohort is an assumption, not a measurement.

Why the base rate transfers — the binding constraint is the grid, not the balance sheet. The natural objection is that AI-era projects carry larger budgets and stronger site control than a speculative wind farm, so they should complete at a higher rate. That objection targets the *developer*; the completion rate is set by the *grid*. The dominant attrition in §3 occurs at the System Impact Study, where a project learns its share of network-upgrade costs — a grid-physics and grid-economics gate (interconnection capacity, transformer and switchgear lead times, transmission build-out) that a larger budget does not remove, only pays into. A well-capitalised data centre still waits behind the same transformer lead times and the same congested interconnection points. The proxy is therefore strongest where the failure is grid-driven (the case here) and weakest where it would be purely developer-economic; that distinction is what the lower rail in §9 rests on, and it is stated rather than assumed.

FERC Order 2023's stricter site-control and deposit rules imply recent cohorts should complete above the historical floor, yet the 2021–2025 cohort has already withdrawn 84% of its MW while building none — too young to confirm either way. State the assumption; give the range.

5. The de-duplication factor on the AI-era cohort (supply side)

On PJM's post-2023 cluster cohort (the AI-surge vintage; 869 non-zero-MW generation projects, 119 GW announced), literal duplicate filings remove only ≈8–19% of MW ($r_1 \approx 0.81$ –0.92, depending on the matching key). The cohort has nonetheless already shed 63% of its MW. The inflation is therefore viability- and withdrawal-driven, not duplication-driven; de-duplication barely moves the headline. The "5–10× phantom filings" prior, read narrowly as literal duplicates, is too high; read broadly as "MW that never get built," it is not refuted — the realized completion in §3 already implies roughly 5× non-completion.

6. What the supply side is — and is not

Supply, not demand: §3–§5 deflate the *generation* side. "Built" counts only In Service (conservative, biasing the rate down). The output is a band, not a point. The analyst value is letting a planner treat an announced generation headline as a scenario, not a baseline.

7. The load side — the data-centre interconnection filter (PJM)

The headline this section addresses is the demand number the supply side does not: the ≈950 TWh-by-2030 trajectory and the "\$850B announced data-centre leases" framing both rest on data-centre *load* requests. Here the method changes for a structural reason, and the change is a gift to reproducibility.

Data centres connect as load, not generation. They do not flow through the generation queue §3 measured, and the large-load process is new (the surge began 2024–25), so there is no matured data-centre load cohort yet — a historical "% energised" rate cannot be computed the way §3 did. But the absence is filled by the operator: PJM performs and publishes the announced-to-deliverable deflation itself, with a documented, reproducible method. This note's contribution on the load side is

therefore explicitly not "we deflate the load." It is to package the operator's own published deflation into a reproducible, cross-region, motive-neutral referee, and to bracket it (\$9).

What PJM publishes. PJM reports three curves for large-load demand:

Curve	2030	Share of submitted	What it is
Submitted (requested)	≈60 GW	100%	large-load demand requests (rises to ≈115 GW by 2046)
Proposed (PJM-accepted, preliminary)	≈38 GW	≈63%	accepted forecast after vetting (≈88 GW by 2046)
Firm only (ESO/CC-backed)	≈32 GW	≈53%	the subset that counts toward capacity / RPM

PJM's own data-centre-specific statement supports "up to ≈30 GW of data-centre growth 2025–2030," noting it is ≈40% of US data centres in 2025 and that it cross-checked FT, Deloitte, BCG, McKinsey and BNEF rather than relying on one source.

The documented haircut mechanics. The vetting is explicit, which is what makes it reproducible rather than a guess. Only projects with an Electric Service Obligation or Construction Commitment are "Firm" and allowed to impact capacity; a default 50% probability is applied to non-firm projects coming online in 3–8 years absent an EDC/LSE factor; a 70% utilization rate and a 70% capacity-to-demand factor are imposed; and a minimum 36-month ramp applies. The worked example in the Implementation document: 2,600 MW requested capacity → 1,800 after probability ($\times \approx 69\%$) → 1,260 MW demand ($\times 70\%$) = ≈48% of requested capacity survives into the forecast. AEP's submission was de-rated by more than 50%.

The final report confirms the direction first-hand. The 2026 Long-Term Load Forecast is lower than the 2025 forecast through 2032, with PJM stating that the firm-vs-non-firm distinction "has brought large load adjustments down in the near-term forecast years." Specific revisions versus the 2025 report: 3rd IA 2026 –2,564 MW (–1.6%); RPM Auction 2028 –4,414 MW (–2.6%); RTEP 2031 –1,630 MW (–0.8%).

Viability gate hardening — a filter, not a solution. A FERC order of 18 December 2025 and PJM's CIFP introduce an Expedited Interconnection Track (effective ≈August 2026) requiring a \$15,000/MW refundable readiness deposit, 100% of network-upgrade cost, and state siting support. A reasonable objection is that these rules make the future queue cleaner than the historical one, so past attrition over-states future attrition. Two points keep that in proportion. First, the rules raise the *barrier to entry*; they do not guarantee *project viability* — a paid deposit moves the failure point earlier (from the study stage to the commitment stage) rather than removing it, so a filtered queue still loses projects to grid costs and build timelines. Second, the existence of the rules is itself operator confirmation that the queue is padded: deposits of this size are not imposed on a credible pipeline. The net effect is to push realized completion toward the *upper* rail of the \$9 bracket over time, which the band already spans.

Sanity rail (not a demand estimate). PJM capacity prices moved \$28.92 → \$269.92 → \$329.17/MW-day (held by a cap) — roughly 10× (IEEFA) — a coarse confirmation that the inflation is material to ratepayers, not a measure of demand.

8. The load side — ERCOT cross-region rail (new in v2)

ERCOT is the cleanest cross-region check for two reasons. First, until the PGRR115 rules it had almost no barrier to entry — by ERCOT's account a speculative phone call could become a formal request — so its queue is maximally padded, the opposite extreme to PJM's gated process. Second, ERCOT publishes a *realized* de-rate (observed peak against requested MW), where PJM's is *applied* (a forward assumption).

Stage	GW	Share of requested	Source
Requested (large-load queue, Dec 2025)	≈226	100%	ERCOT System Planning Update Dec 2025 (≈233 GW at the 9 Dec board)
Credible 2030 data-centre scenario	≈35	≈15%	Aurora Energy Research (ERCOT-commissioned resource-adequacy study)
Energized to date (18 Nov 2025)	5,302 MW	≈2%	ERCOT Large Load Interconnection Process Q&A, 24 Dec 2025

The queue is ≈70–77% data centres and grew from ≈63 GW at the end of 2024. ERCOT's own characterisation is that most of it "will not materialize."

Three quantities that must not be conflated. ERCOT's data separates three different things, and the analysis turns on keeping them apart:

- **Request survival** — how much of the requested queue survives vetting into the credible forecast. ERCOT's commissioned scenario puts deliverable 2030 data-centre load near ≈35 GW against a ≈226 GW queue, and ERCOT will, after 2026, count only executed-agreement load. This is the de-rate of the *request*.
- **Realized utilization (performance)** — for data-centre sites that actually got built (2022–2024 in-service dates), the average per-site peak consumption is 49.8% of the requested MW (ERCOT Long-Term Load Forecast 2025, non-crypto additions). A related ERCOT figure (via the WECC large-loads risk assessment, Figure 2.2) shows only 58–78% of requested transmission service is used on peak within a year. This is a *performance* measure, conditional on the project existing.
- **Energized to date** — 5,302 MW (≈2% of ERCOT's large-load tracking) had been observed energized as of 18 November 2025. This is an *age* figure, not a failure rate: the surge began in 2024–25 and a large load takes years to build and ramp, so a low energised share is expected and says nothing about eventual completion. Reading the 2% as "98% will fail" is the misread the three-way split is designed to prevent.

The forecast gate (ERCOT's analog of PJM's Firm/ESO-CC). PGRR115 / NPRR1234, effective 15 December 2025, require executing all agreements, notice to proceed, and payment of financial

obligations before approval to energize; going forward, only executed-agreement loads enter the large-load forecast. Texas SB6 adds upfront study fees and duplicate-request disclosure, with PUCT rulemaking due December 2026.

The cross-region finding. Two independent operators, using different methods, reach the same structural conclusion: the requested queue is a scenario, not a baseline, and roughly half survives the first credible screen.

Operator	Method	First-screen survival
PJM	forward de-rate (probability $\approx 0.69 \times$ utilization 0.70)	$\approx 48\%$ of requested capacity
ERCOT	realized de-rate (per-site peak $\div$ requested MW)	49.8%

The honest caveat, stated rather than glossed: these two $\approx 50\%$ figures are *constructed differently*. PJM's bundles a completion-probability factor; ERCOT's is conditional on the project having been built (pure peak-vs-requested utilization, excluding non-completion). ERCOT's 49.8% is therefore independent empirical support for the order of magnitude of PJM's applied haircut, not a like-for-like identity. The robust claim is structural: both operators find requested far exceeds deliverable, both haircut by roughly half at the first credible step, and both now gate the forecast on an executed agreement.

9. The two-rail energisation bracket (formalised)

With no matured data-centre load cohort, the deliverable rate cannot be measured directly; it is bracketed by two rails, and the gap between them is the finding.

- **Upper rail (operator-accepted into the forecast).** PJM: $\approx 63\%$ of 2030 submitted survives vetting (Firm-only $\approx 53\%$). ERCOT: a realized $\approx 50\%$ per-site. Call the upper rail $\approx 50\text{--}63\%$.
- **Lower rail (energisation proxy).** No load cohort has resolved, so the floor is borrowed from the supply side: generation-queue historical completion $\approx 20\text{--}22\%$ (§3 / LBNL). This transfers on the grid-physics channel argued in §4 — generation and load clear the *same* interconnection queue, network-upgrade-cost gate, and equipment lead times — so it is a defensible floor where the binding constraint is the grid, and a labelled proxy, not a measured load-completion rate, where it is not.

The deliverable band for announced data-centre demand is $\approx 20\%$ (energisation floor) to $\approx 50\text{--}63\%$ (operator-accepted ceiling). The band deliberately spans the regime change: the lower rail is the old, lightly-gated queue's completion; the upper rail is the operators' newly-vetted acceptance. By the operators' own primary vetting, roughly half of the announced large-load ask survives to the forecast, and historical energisation is lower again. Announced is a scenario, not a baseline.

ERCOT's $\approx 2\%$ energized-to-date sits below even the floor because its cohort is too young to have built out — the age artefact of §8, not a completion rate, and a reminder that the lower rail is conservative.

Who this serves. The value is to the planner, not to a market view. Treating a requested queue as firm leads to over-procurement of generation and transmission; the cost of that over-build lands on ratepayers through capacity-price spikes of the kind the \$7 sanity rail records (a roughly 10× move). Deflating the headline to a band lets a planner size investment to deliverable load rather than to a contaminated request, which is a ratepayer-protection argument as much as an analytical one.

10. What v2 is — and is not

- **The contribution is the packaging, not the deflation.** On the load side the operators deflate and publish; restating that would add nothing. What is absent — and what this provides — is a reproducible, cross-region, motive-neutral referee that places PJM and ERCOT side by side on a common frame, brackets the accepted figure against a realized-completion floor, and names every assumption. The same empty ground the supply-side anchor filled, on the demand side.
- **Not a crash call.** Method-first and agnostic on whether or when a financial correction lands. The robust-vs-contingent split holds either way.
- **"Avoided," not "saved."** Deflated demand is capacity never built, or built and idled (stranded) — not efficiency saved.
- **Two named load-bearing assumptions.** Supply side: cohort comparability, carried by the grid-as-binding-constraint argument (§4). Load side: that the operators' published de-rates and the executed-agreement gate transfer to the energised outcome (the upper rail is acceptance, not energisation; the lower rail is a generation proxy that holds on the grid-physics channel). Both are stated; the band spans them.
- **OOM, not precision.** The output is a band with its softest inputs marked, in both regions.

11. How the bot paper slots in

The bot-energy paper is the proof-of-method: the deflation discipline was built and validated on a clean, low-stakes case — where the author pushed his own bound up against his interest — before being pointed at the numbers that matter. The sequencing demonstrates the trait the work is for: not inflating even when inflating makes the better headline. v0.1 applied it to the supply side; v2 extends it to the demand side and to a second region, and is careful to claim only the packaging, since the operators own the deflation. [1]

Sources. Supply side: PJM Serial / Cycle Service Request Status (queue data, status, MW, milestones); LBNL "Queued Up" (national historical completion ≈14–20%); Exelon pipeline disclosures (≈22% of company pipeline); FERC Order 2023. Load side, PJM: Load Adjustment Request Implementation (1 Jul 2025); Load Adjustment Requests Summary for the 2026 Load Forecast — Preliminary (24 Nov 2025); 2026 Long-Term Load Forecast Report (posted 14 Jan 2026); FERC order 18 Dec 2025; IEEFA (capacity-price sanity rail). Load side, ERCOT: System Planning and Weatherization Update (Dec 2025); Large Load Interconnection Process Q&A (24 Dec 2025); Long-Term Load Forecast 2025 (the 49.8% per-site figure); NPRR1234 / PGRR115 market notices (forecast gate, eff. 15 Dec 2025); Aurora Energy Research resource-adequacy study for ERCOT (≈35 GW 2030 scenario); WECC, Assessment of Large Load Interconnection Risks (2025,

Fig. 2.2); Belfer Center, Data Centers and Large-Scale Electric Growth: the Virginia and Texas Experiences (2026); Utility Dive and Latitude Media (queue-growth reporting citing ERCOT board figures); Texas SB6 (2025). Demand-trajectory framing: IEA, Key Questions on Energy and AI and Electricity 2026 (the 485 → 950 TWh trajectory and bottleneck caveats). Figures are dated and subject to revision; supply-side figures were confirmed by re-running the corrected script on 2026-06-15, and load-side figures are published operator aggregates. This note is independent analysis, not investment advice. Reproducible via `reproduce.py` (supply side, against a locally-downloaded PJM export) and the published load-side aggregates (`dc\_funnel.csv`, `ercot\_funnel.csv`).

[1] Conflict of interest. This paper was prepared with the assistance of an Anthropic model (Claude Opus 4.8). Anthropic is a party to the AI data-centre buildout whose demand this paper deflates, and appears among the private-capital announcements that motivate the demand headline; deflating AI power demand is not unambiguously for or against that interest, but the assistance is disclosed for completeness. No third party reviewed, funded, or directed this work.

